Additive Representations by Primitive Dyck Numbers

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Abstract

Reading a Dyck word as a binary integer, with 1 as an opening bracket and 0 as a closing bracket, gives a totally balanced number. We study the subset arising from primitive Dyck words, whose running balance returns to zero only at the end. We determine the least number of primitive Dyck numbers needed to represent every positive even integer. Among positive even integers, 46 alone requires exactly eight summands, and the integers 34, 44, 98, 154, 198, 202, 206, 838, 842, and 846 require seven. Every other positive even integer is a sum of at most six primitive Dyck numbers. In particular, 848 is the sharp threshold beyond which six summands always suffice. The proof combines a regular sublanguage with a self-similar closure property of the full primitive family after division by 4.

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1 Introduction

Every string of balanced parentheses can be written over the alphabet $\{1, 0\}$, with 1 for an opening bracket and 0 for a closing one, and so read as a binary integer. Which integers arise this way, and how additively rich is the set they form? We take up a natural variant of that question.

A *Dyck word* is a word $w \in \{1, 0\}^*$ with equally many 1's and 0's in which every prefix has at least as many 1's as 0's; thus 1 opens a parenthesis and 0 closes it. A nonempty Dyck word is *primitive* if none of its proper nonempty prefixes is itself balanced—equivalently, if the associated lattice path returns to height zero for the first time only at its endpoint. Such paths are also commonly called *irreducible* or *indecomposable* Dyck paths. Let $\mathcal{D}_{\mathrm{P}}$ denote the language of primitive Dyck words. Ordered by length and then lexicographically, its first eighteen elements are

$$\begin{gathered}10,\ 1100,\ 110100,\ 111000,\ 11010100,\ 11011000,\\ 11100100,\ 11101000,\ 11110000,\ 1101010100,\ 1101011000,\\ 1101100100,\ 1101101000,\ 1101110000,\ 1110010100,\\ 1110011000,\ 1110100100,\ 1110101000.\end{gathered}$$

For a word w over the base-m digits, let $[w]_m$ denote the integer it represents in base m. Adjoining 0 for convenience, set

$$\mathcal{N}_{\mathrm{pD}} = \{0\} \cup \{[w]_2 : w \in \mathcal{D}_{\mathrm{P}}\}.$$

The first eighteen positive elements are

$$\begin{gathered}2, 12, 52, 56, 212, 216, 228, 232, 240,\\ 852, 856, 868, 872, 880, 916, 920, 932, 936;\end{gathered}$$

this is OEIS sequence A057547 [?]. Every positive member of $\mathcal{N}_{\mathrm{pD}}$ is even, since every nonempty Dyck word ends in 0. Thus only even target integers can be represented by sums from $\mathcal{N}_{\mathrm{pD}}$.

The following sharper congruence observation will be used repeatedly.

Lemma 1. *The number* 2 *is the only positive primitive Dyck number congruent to* 2 *modulo* 4. *Every other positive primitive Dyck number is divisible by* 4.

Proof. Every nonempty Dyck word ends in 0. A primitive Dyck word of length at least four must in fact end in 00: if it ended in 10, the path would already have returned to height zero just before its last two letters. The word 10 represents 2, and every longer primitive word therefore represents a multiple of 4. □

Bell, Lidbetter, and Shallit studied the larger set obtained from *all* Dyck words, the *totally balanced numbers* (OEIS sequence A014486 [?]). Using regular underapproximations and automata, they proved that every even integer except

$$8,\ 18,\ 28,\ 38,\ 40,\ 82,\ 166$$

is a sum of at most three totally balanced numbers, while two summands do not suffice asymptotically [**?**, Thm. 14].

The primitive restriction is not a cosmetic variant of that result. Every nonempty Dyck word factors uniquely as a concatenation of primitive Dyck words, whereas here each summand must consist of a single factor. Consequently, a representation by unrestricted Dyck numbers does not in general yield a representation by the same number of primitive Dyck numbers.

Digit-defined additive bases have also been studied by automata-theoretic methods. Bell, Hare, and Shallit characterized automatic sets that form additive bases and gave an effective procedure for determining their orders [**?**]. Related decision procedures have yielded sharp additive results for binary palindromes [**?**] and binary squares [**?**]. The language of primitive Dyck words is not regular, so those general finite-automaton methods do not directly settle the present problem.

Our proof differs at the structural point where regular approximation alone ceases to suffice. A regular sublanguage settles the congruence class 0 modulo 4, but the class 2 modulo 4 is handled by a self-similar closure property of the full primitive family. This leads not only to an eventual bound but to the complete exceptional set, the optimal uniform bound for all even integers, and the sharp eventual threshold.

Within the Dyck words of length $2n$, the proportion that are primitive tends to $1/4$ as $n \to \infty$. Indeed, there are C_n Dyck words of length $2n$, and every primitive one is uniquely of the form $1u0$ with u a Dyck word of length $2n-2$; so there are C_{n-1} primitive words of length $2n$. From the Catalan formula $C_n = \frac{1}{n+1}\binom{2n}{n}$,

$$\frac{C_{n-1}}{C_n} = \frac{n+1}{2(2n-1)} \longrightarrow \frac{1}{4}.$$

Its arithmetic effect, however, is striking. Although all sufficiently large even integers need only three unrestricted Dyck summands, the primitive family has a small but nontrivial exceptional set. We show that precisely eleven positive even integers fail to be sums of at most six primitive Dyck numbers. Ten require seven summands, while 46 alone requires eight. All eleven exceptions lie below 848.

The need for eight summands is entirely local: below 48 the only positive primitive Dyck numbers are 2 and 12. The eventual six-summand bound, on the other hand, comes from two self-similar families. A regular subfamily handles multiples of 4, while the full primitive family after division by 4 handles the remaining congruence class.

For a positive even integer N, let $r(N)$ be the least number of positive primitive Dyck numbers whose sum is N, and define the central integer sequence of this paper by

$$a(n) = r(2n), \qquad n \geq 1.$$

This is OEIS sequence A395858 [**?**]. Its first fifty terms are

$$1, 2, 3, 4, 5, 1, 2, 3, 4, 5, 6, 2, 3, 4, 5, 6, 7, 3, 4, 5, 6, 7, 8, 4, 5,$$
$$1, 2, 1, 2, 3, 4, 2, 3, 2, 3, 4, 5, 3, 4, 3, 4, 5, 6, 4, 5, 4, 5, 6, 7, 5.$$

The displayed terms and the finite certificates used below are reproducible from the accompanying Python programs.

Theorem 2. *The values of $r(N)$ exceeding six are exactly as follows:*

$$r(46) = 8,$$

and

$$r(N) = 7 \quad \Longleftrightarrow \quad N \in \{34, 44, 98, 154, 198, 202, 206, 838, 842, 846\}.$$

Every other positive even integer satisfies $r(N) \leq 6$. Consequently, every even integer $N \geq 848$ is a sum of at most six primitive Dyck numbers.

Equivalently, in terms of the sequence $a(n) = r(2n)$,

$$\max_{n \geq 1} a(n) = 8, \qquad a(n) = 8 \quad \Longleftrightarrow \quad n = 23,$$

and

$$a(n) = 7 \quad \Longleftrightarrow \quad n \in \{17, 22, 49, 77, 99, 101, 103, 419, 421, 423\}.$$

Thus the main theorem gives a complete determination of all terms of $a(n)$ exceeding six, rather than only an eventual estimate.

The exceptional value 46 already explains why the uniform bound can never be smaller than eight.

Proposition 3. *The integer* 46 *requires exactly eight positive primitive Dyck numbers: it is a sum of eight, but it is not a sum of at most seven.*

Proof. Below 48 the only positive primitive Dyck numbers are 2 and 12. Indeed, the primitive words of length two and four are 10 and 1100, whereas every primitive word of length at least six begins with 11 and therefore represents an integer at least $[110000]_2 = 48$. Thus any representation of 46 has the form

$$46 = 12a + 2b, \qquad a, b \geq 0.$$

Equivalently, $6a + b = 23$. Since $12 \cdot 4 > 46$, we have $a \leq 3$, and hence the number of summands satisfies

$$a + b = 23 - 5a \geq 8.$$

Equality is attained by

$$46 = 12 + 12 + 12 + 2 + 2 + 2 + 2 + 2.$$

□

For example,

$$2026 = 2 + 240 + 852 + 932 = 2 + 216 + 872 + 936.$$

The seven distinct integers appearing in these two representations all have primitive Dyck binary expansions.

The sumset notation used below is standard in additive-basis theory [**?**, Chap. 1]. For a nonnegative integer k, let kX denote the set of sums of k not necessarily distinct members of X, with $0X = \{0\}$, and for a scalar c write $c \cdot X = \{cx : x \in X\}$ for the dilation of X.

For $A \subseteq 2\mathbb{N}_0$, we call A a *relative additive basis of order at most h for* $2\mathbb{N}_0$ if every element of $2\mathbb{N}_0$ is a sum of at most h elements of A. Its relative order is *exactly* h if h is the least integer with this property. We similarly call A a *relative asymptotic additive basis of order at most h* if every sufficiently large element of $2\mathbb{N}_0$ is a sum of at most h elements of A.

2 A regular family inside the primitive words

A simple regular pattern runs through the primitive Dyck words. For example, the binary expansion of 868 factors as

$$\mathrm{bin}(868) = 1101100100 = 11\,01\,10\,01\,00 \in 11(01 \mid 10)^*00.$$

Pairing the binary digits turns the initial block 11 into the base-4 digit 3, the final block 00 into 0, and each middle block 01 or 10 into 1 or 2. This suggests that sums of a fixed number of such base-4 integers might fill long intervals. The mixed sumset lemma below makes the idea precise.

Let

$$\mathcal{E} = \{0\} \cup \big\{[3\varepsilon_1 \cdots \varepsilon_k]_4 : k \geq 0,\ \varepsilon_i \in \{1,2\}\big\}.$$

When $k = 0$ the string $\varepsilon_1 \cdots \varepsilon_k$ is empty, so the corresponding element is $[3]_4 = 3$. Multiplying by 4 appends a base-4 zero. Replacing each base-4 digit by its two binary digits,

$$\begin{aligned} 3_{(4)} &\longleftrightarrow 11_{(2)}, & 1_{(4)} &\longleftrightarrow 01_{(2)}, \\ 2_{(4)} &\longleftrightarrow 10_{(2)}, & 0_{(4)} &\longleftrightarrow 00_{(2)}. \end{aligned}$$

we find that every positive element of $4 \cdot \mathcal{E}$ has binary expansion in $11(01 \mid 10)^*00$. After the initial 11 the path stands at height 2; each middle block 01 or 10 returns it to height 2; and the final 00 brings it to height zero for the first time. Thus every positive element of $4 \cdot \mathcal{E}$ is a primitive Dyck number, and, since $0 \in \mathcal{N}_{\mathrm{pD}}$,

$$4 \cdot \mathcal{E} \subseteq \mathcal{N}_{\mathrm{pD}}. \tag{1}$$

The crux is that six elements of $\mathcal{E}$ already cover every integer from 25 on.

Proposition 4. *Every integer $n \geq 25$ belongs to $6\mathcal{E}$.*

We prove this by a finite-digit interval argument. The role of the auxiliary sets below is worth explaining first. When an m-digit element of $\mathcal{E}_m$ is split into its leading digit and its lower digits, the lower $m-1$ digits form a base-4 correction whose digits lie in $\{0,1\}$. The set Δ_m records exactly these corrections. When some summands in a six-term sum use a new leading digit, the remaining lower-digit problem becomes a mixed sum of copies of Δ_{m-1} and $\mathcal{E}_{m-1}$. The mixed sumsets $S_{t,m}$ are designed to keep track of all these possibilities simultaneously.

Set $u_0 = 0$ and, for $m \geq 1$,

$$u_m = [\underbrace{11 \cdots 1}_{m \text{ digits}}]_4 = 1 + 4 + \cdots + 4^{m-1} = \frac{4^m - 1}{3}, \tag{2}$$

so that

$$\begin{aligned} 4^{m-1} &= 3u_{m-1} + 1 \quad (m \geq 1), \\ u_{m-1} &= 4u_{m-2} + 1 \quad (m \geq 2). \end{aligned} \tag{3}$$

Define the correction set

$$\Delta_m = \Big\{[\delta_{m-1}\cdots\delta_0]_4 : \delta_j \in \{0,1\}\Big\} = \Big\{\textstyle\sum_{j=0}^{m-1}\delta_j 4^j : \delta_j \in \{0,1\}\Big\},$$

so that $\max\Delta_m = u_m$. Let $\mathcal{E}_m$ be the finite subset of $\mathcal{E}$ consisting of 0 together with the elements having at most m base-4 digits. For example,

$$\begin{aligned}\Delta_2 &= \{[00]_4, [01]_4, [10]_4, [11]_4\} = \{0,1,4,5\},\\ \mathcal{E}_2 &= \{0, [3]_4, [31]_4, [32]_4\} = \{0,3,13,14\}.\end{aligned}$$

For $0 \le t \le 6$ set

$$S_{t,m} = t\Delta_m + (6-t)\mathcal{E}_m.$$

The largest element of $\mathcal{E}_m$ is

$$\max\mathcal{E}_m = [3\,\underbrace{22\cdots2}_{m-1\text{ digits}}\,]_4 = 3\cdot 4^{m-1} + 2u_{m-1},$$

and for $m \ge 2$ equation (**??**) gives

$$\max\mathcal{E}_{m-1} = 11u_{m-2} + 3 = \frac{11u_{m-1}+1}{4}. \tag{4}$$

Since $\max\Delta_m = u_m$,

$$\max S_{t,m} = tu_m + (6-t)\max\mathcal{E}_m. \tag{5}$$

Moreover $\max\mathcal{E}_m > u_m$, so $\max S_{t,m}$ decreases with t. Throughout, $[a,b]$ denotes the integer interval $\{a, a+1, \ldots, b\}$.

Lemma 5. *For every $m \ge 2$ and every $2 \le t \le 6$,*

$$S_{t,m} = [0, \max S_{t,m}], \tag{6}$$

whereas

$$S_{1,m} = [0, \max S_{1,m}] \setminus \{2\}. \tag{7}$$

Moreover,

$$[25, \ell_m] \subseteq 6\mathcal{E}_m, \qquad \ell_m = \frac{33}{8}\,4^m - 4. \tag{8}$$

Proof. We first prove (**??**) and (**??**) in the base case $m = 2$. Recall that

$$\Delta_2 = \{0, 1, 4, 5\}, \qquad \mathcal{E}_2 = \{0, 3, 13, 14\}.$$

Direct calculation gives $2\Delta_2 = \{0, 1, 2, 4, 5, 6, 8, 9, 10\}$, and adding one more copy of Δ_2 fills every remaining gap, so $3\Delta_2 = [0, 15]$ and hence

$$t\Delta_2 = [0, 5t] \qquad (3 \le t \le 6).$$

On the other hand,

$$\begin{aligned}
2\mathcal{E}_2 &= \{0, 3, 6, 13, 14, 16, 17, 26, 27, 28\},\\
3\mathcal{E}_2 &= \{0, 3, 6, 9\} \cup [13, 14] \cup [16, 17] \cup [19, 20]\\
&\qquad \cup [26, 31] \cup [39, 42],\\
4\mathcal{E}_2 &= \{0, 3, 6, 9\} \cup [12, 14] \cup [16, 17] \cup [19, 20]\\
&\qquad \cup [22, 23] \cup [26, 34] \cup [39, 45] \cup [52, 56],\\
5\mathcal{E}_2 &= \{0, 3, 6, 9\} \cup [12, 17] \cup [19, 20] \cup [22, 23]\\
&\qquad \cup [25, 37] \cup [39, 48] \cup [52, 59] \cup [65, 70].
\end{aligned}$$

Adding Δ_2 to $5\mathcal{E}_2$, and $2\Delta_2$ to $4\mathcal{E}_2$, gives

$$S_{1,2} = \{0, 1\} \cup [3, 75], \qquad S_{2,2} = [0, 66].$$

Similarly,

$$\begin{aligned}
S_{3,2} &= 3\Delta_2 + 3\mathcal{E}_2 = [0, 15] + 3\mathcal{E}_2 = [0, 57],\\
S_{4,2} &= 4\Delta_2 + 2\mathcal{E}_2 = [0, 20] + 2\mathcal{E}_2 = [0, 48],\\
S_{5,2} &= 5\Delta_2 + \mathcal{E}_2 = [0, 25] + \mathcal{E}_2 = [0, 39],\\
S_{6,2} &= 6\Delta_2 = [0, 30].
\end{aligned}$$

Collecting these,

t	$S_{t,2}$
1	$[0, 75] \setminus \{2\}$
2	$[0, 66]$
3	$[0, 57]$
4	$[0, 48]$
5	$[0, 39]$
6	$[0, 30]$

which proves (**??**) and (**??**) for $m = 2$. For (**??**) we must check $[25, \ell_2] = [25, 62] \subseteq 6\mathcal{E}_2$. The intervals $[26, 28]$, $[39, 42]$, and $[52, 56]$ are the sums of two, three, and four members of $\{13, 14\}$, respectively, so

$$\begin{aligned}
25 &= 13 + 3 + 3 + 3 + 3 + 0, \\
[26, 40] &= \{26, 27, 28\} + 3 \cdot \{0, 1, 2, 3, 4\}, \\
[39, 51] &= [39, 42] + 3 \cdot \{0, 1, 2, 3\}, \\
[52, 62] &= [52, 56] + 3 \cdot \{0, 1, 2\}.
\end{aligned}$$

This completes the base case. The lower endpoint is sharp: $24 \notin 6\mathcal{E}$, since the only positive elements of $\mathcal{E}$ below 25 are $3, 13$, and 14, and no sum of at most six of these is 24.

Now let $m \geq 3$ and assume that (**??**)–(**??**) hold at level $m - 1$. To lighten the notation during this step, write

$$u = u_{m-1}, \qquad q = q_m, \qquad M = \max \mathcal{E}_{m-1}.$$

Then

$$4^{m-1} = 3u + 1, \qquad q = 10u + 3, \qquad M = \frac{11u + 1}{4}.$$

Here

$$q = q_m = [3\,\underbrace{11 \cdots 1}_{m-1 \text{ digits}}]_4 = 3 \cdot 4^{m-1} + u = 10u + 3, \tag{9}$$

and we shall repeatedly use

$$2 \cdot 4^{m-1} + 4u = q - 1. \tag{10}$$

Splitting according to the leading base-4 digit gives

$$\Delta_m = \Delta_{m-1} \cup (4^{m-1} + \Delta_{m-1}), \qquad \mathcal{E}_m = \mathcal{E}_{m-1} \cup (q + \Delta_{m-1}).$$

Indeed, the new leading digit of an element of Δ_m is 0 or 1, whereas a new m-digit element of $\mathcal{E}_m$ has leading digit 3 followed by a lower $(m-1)$-digit correction from Δ_{m-1}. If s of the t copies of Δ_m use their shifted part and v of the $6 - t$ copies of $\mathcal{E}_m$ use their new m-digit part, then

$$S_{t,m} = \bigcup_{v=0}^{6-t} \bigcup_{s=0}^{t} \big(s \cdot 4^{m-1} + vq + S_{t+v,m-1}\big). \tag{11}$$

Thus, for fixed v, the parameter s produces $t+1$ translates by multiples of 4^{m-1}; increasing v then moves to the next large block:

$$\underbrace{[vq,\ \cdots]}_{v\text{-block}} \qquad \underbrace{[(v+1)q,\ \cdots]}_{(v+1)\text{-block}}.$$

The rest of the proof checks that neighboring translates and neighboring v-blocks overlap.

We first treat $t \geq 2$. The induction hypothesis (**??**) gives $S_{t+v,m-1} = [0, \max S_{t+v,m-1}]$ for every value of v in (**??**). For fixed v, consecutive values of s shift this interval by 4^{m-1}. Since $\max S_{j,m-1}$ decreases with j,

$$\max S_{t+v,m-1} \geq \max S_{6,m-1} = 6u \geq 3u = 4^{m-1} - 1,$$

so these $t+1$ translates overlap and form

$$[vq,\ vq + t \cdot 4^{m-1} + \max S_{t+v,m-1}].$$

The blocks for consecutive values of v also overlap. Whenever a next block exists we have $t + v \leq 5$, and since $t \geq 2$,

$$\begin{aligned} t \cdot 4^{m-1} + \max S_{t+v,m-1} &\geq 2 \cdot 4^{m-1} + \max S_{5,m-1} \\ &= 2 \cdot 4^{m-1} + 5u + M \\ &= (2 \cdot 4^{m-1} + 4u) + (u + M) \\ &\geq q - 1, \end{aligned}$$

by (**??**). Hence the union in (**??**) is the full integer interval from 0 to its largest element $\max S_{t,m}$. This proves (**??**) at level m.

For $t = 1$ the first block retains the single missing integer 2. For each fixed $v \geq 1$, the two s-translates involve $S_{1+v,m-1}$ with $1+v \geq 2$; they overlap because

$$\max S_{1+v,m-1} \geq \max S_{6,m-1} = 6u \geq 4^{m-1} - 1.$$

Every later v-block is therefore an interval. Since $\max S_{j,m-1}$ decreases with j, it suffices to check the last pair of consecutive v-blocks. We have

$$\begin{aligned} 4^{m-1} + \max S_{5,m-1} - (q-1) &= M - 2u - 1 \\ &= \frac{11u+1}{4} - 2u - 1 = \frac{3u-3}{4} \geq 0, \end{aligned}$$

so consecutive v-blocks overlap. Finally,

$$\begin{aligned}\max S_{1,m-1} - (4^{m-1} + 2) &= u + 5M - 4^{m-1} - 2 \\ &= \frac{47u - 7}{4} \geq 0.\end{aligned}$$

Thus, within the first v-block, the unshifted long interval reaches $4^{m-1} + 2$ and joins the next s-translate without a gap. Hence

$$S_{1,m} = [0, \max S_{1,m}] \setminus \{2\},$$

which proves (**??**) at level m.

It remains to propagate the interval inside $6\mathcal{E}_m$. Taking $t = 0$ in (**??**),

$$6\mathcal{E}_m = \bigcup_{v=0}^{6} \big(vq + S_{v,m-1}\big).$$

The $v = 0$ block is $S_{0,m-1} = 6\mathcal{E}_{m-1}$, which contains $[25, \ell_{m-1}]$ by the induction hypothesis. Since $m \geq 3$ we have $u \geq 5$, and

$$\ell_{m-1} - (q + 2) = \frac{19u - 39}{8} \geq 0.$$

The $v = 1$ block, namely $q + S_{1,m-1}$, contains $\{q, q+1\} \cup [q+3, q+\max S_{1,m-1}]$ by the induction hypothesis. Because the preceding interval reaches through $q + 2$, the two join without a gap. To join the $v = j$ block to the $v = j + 1$ block for $j = 1, 2, 3$, it is enough to have

$$\max S_{j,m-1} \geq q - 1.$$

Since $\max S_{j,m-1}$ decreases with j, the most restrictive case is $j = 3$; hence it suffices to check

$$\begin{aligned}\max S_{3,m-1} - (q - 1) &= 3u + 3M - (10u + 2) \\ &= \frac{5u - 5}{4} \geq 0.\end{aligned}$$

Thus $\max S_{1,m-1}$, $\max S_{2,m-1}$, $\max S_{3,m-1} \geq q - 1$, and the blocks $v = 1, 2, 3, 4$ join in turn and cover through

$$4q + \max S_{4,m-1} = \frac{33}{8}\, 4^m - 4 = \ell_m.$$

The remaining blocks $v = 5, 6$ may begin after a gap, but they are not needed for the interval propagated here. This establishes (**??**)–(**??**) at level m and completes the induction. $\square$

Since $\ell_m \to \infty$ and $\mathcal{E} = \bigcup_{m\geq 1} \mathcal{E}_m$, Proposition **??** follows at once.

The preceding proposition will handle the congruence class 0 modulo 4: every multiple of 4 at least 100 is a sum of at most six primitive Dyck numbers.

3 The full primitive family after division by four

The regular family $\mathcal{E}$ is sufficient for multiples of 4, but it does not by itself provide the five-summand interval needed below. For the other congruence class we instead use the full primitive family. Set

$$\left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\right)_{\geq 12} = \{v/4 : v \in \mathcal{N}_{\mathrm{pD}},\ v \geq 12\}.$$

Thus

$$\mathcal{N}_{\mathrm{pD}} = \{0, 2\} \cup (4 \cdot \left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\right)_{\geq 12}).$$

For a set X of integers and $k \geq 0$, write

$$F_k(X) = \bigcup_{j=0}^{k} jX$$

for the set of sums of at most k elements of X.

The full family has a simple self-similar closure property.

Lemma 6. *If $p \in \left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\right)_{\geq 12}$, then $4p+1$ and $4p+2$ also belong to $\left(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\right)_{\geq 12}$.*

Proof. Write the binary expansion of the primitive Dyck number $4p$ as $u00$. Immediately after reading u, the associated path has height 2. The binary expansions of $4(4p+1)$ and $4(4p+2)$ are, respectively,

$$u0100 \qquad \text{and} \qquad u1000.$$

The inserted blocks 01 and 10 both begin and end at height 2, and their intermediate heights are positive. Thus both new paths remain strictly above zero until their final symbol and return to zero only at the end. The two displayed words are therefore primitive Dyck words, which proves the claim. $\square$

We next establish a five-summand interval for $\left(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\right)_{\geq 12}$. The finite base uses the two subsets

$$L = \{3, 13, 14, 53, 54, 57, 58, 60\} \subseteq \left(\tfrac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\right)_{\geq 12}$$

and

$$H = \{213, 214, 217, 218, 220, 229, 230, 233, 234, 236,\\ 241, 242, 244, 248\} \subseteq \left(\tfrac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\right)_{\geq 12}.$$

The following tables display all the corresponding primitive Dyck numbers. In each entry the binary word $[4h]_2$ is primitive, and hence $h \in \left(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\right)_{\geq 12}$.

$4L$

h	$[4h]_2$	h	$[4h]_2$
3	1100	53	11010100
13	110100	54	11011000
14	111000	57	11100100
58	11101000	60	11110000

$4H$

h	$[4h]_2$	h	$[4h]_2$
213	1101010100	233	1110100100
214	1101011000	234	1110101000
217	1101100100	236	1110110000
218	1101101000	241	1111000100
220	1101110000	242	1111001000
229	1110010100	244	1111010000
230	1110011000	248	1111100000

The finite computations can be reproduced using the Python program `verify_certificates_for_paper.py`, archived at commit `bc0211ccb478d79969f8ce6b5fe465e5f3fd9cf5`: `https://github.com/growupkuriyama-hub/lean_cfg_project/blob/bc0211ccb478d79969f8ce6b5fe465e5f3fd9cf5/LeanCfgProject/PrimDyck/verify_certificates_for_paper.py`. The same program is included as a supplementary file, together with a README giving the execution command and the expected output.

The five statements in Lemma **??** are inclusion certificates: the program checks that each displayed interval is contained in the corresponding sumset, and does not claim that the sumset equals that interval. By contrast, for Lemma **??** the program computes the relevant sets truncated to $[0, 211]$ exactly and checks equality with the displayed unions of intervals.

Lemma 7 (Finite interval certificate)**.** *The following inclusions hold:*

sumset	*integer interval contained*
$5L$	$[215, 254]$
$H + 4L$	$[245, 486]$
$2H + 3L$	$[439, 674]$
$3H + 2L$	$[645, 862]$
$4H + L$	$[855, 1046]$.

Proof. Put

$$A = \{53, 54, 57, 58, 60\}, \qquad B = \{3, 13, 14\}.$$

Successive additions give

$$2A = [106, 108] \cup [110, 118] \cup \{120\},$$

$$3A = [159, 178] \cup \{180\}, \qquad 4A = [212, 238] \cup \{240\}.$$

Hence $4A + B = [215, 254]$, proving the first line.

For the second line, repeated addition of L gives

$$\begin{aligned} 4L \supseteq\ & [32, 34] \cup [42, 45] \cup [52, 56] \cup [82, 102] \\ & \cup [116, 148] \cup [162, 194] \cup [212, 238]. \end{aligned}$$

The three short intervals give

$$\begin{aligned} H + [32, 34] &\supseteq [245, 254] \cup [261, 270] \cup [273, 278] \cup [280, 282], \\ H + [42, 45] &\supseteq [255, 265] \cup [271, 281] \cup [283, 293], \\ H + [52, 56] &\supseteq [265, 276] \cup [281, 304]. \end{aligned}$$

Their union contains $[245, 304]$. Consecutive elements of H differ by at most 9. Therefore, if $I = [a, b]$ with $b - a \geq 8$, then $H + I = [213 + a, 248 + b]$. Applying this to the four remaining long intervals gives

$$[295, 350], \quad [329, 396], \quad [375, 442], \quad [425, 486],$$

which overlap and complete the proof that $[245, 486] \subseteq H + 4L$.

For the third line, the following contained intervals suffice:

$$2H \supseteq [426, 428] \cup [430, 438] \cup [446, 478] \cup \{496\},$$

$$\begin{aligned} 3L \supseteq \{9\} \cup [19, 20] \cup [39, 42] \cup [73, 77] \cup [109, 111] \\ \cup [113, 134] \cup [159, 178] \cup \{180\}. \end{aligned}$$

The first two components of $2H$ give

$$[430, 438] + 9 = [439, 447], \quad [426, 428] + [19, 20] = [445, 448],$$

and

$$[430, 438] + [19, 20] = [449, 458].$$

Adding $[446, 478]$ to, successively,

$$\begin{gathered} \{9\},\ [39, 42],\ [73, 77],\ [109, 111], \\ [113, 134],\ [159, 178],\ \{180\} \end{gathered}$$

produces overlapping intervals covering $[455, 658]$. Finally,

$$496 + [159, 178] = [655, 674].$$

Together these intervals prove $[439, 674] \subseteq 2H + 3L$.

For the fourth line, use

$$3H \supseteq [639, 734] \cup \{744\}, \qquad 2L \supseteq \{6\} \cup [70, 74] \cup [110, 118] \cup \{120\}.$$

The four interval sums

$$[645, 740], \quad [709, 808], \quad [759, 854], \quad [854, 862]$$

cover $[645, 862]$. For the last line, use

$$4H \supseteq [852, 982] \cup [984, 986], \qquad \{3, 57, 58, 60\} \subseteq L.$$

The intervals

$$[855, 985], \quad [912, 1042], \quad [1041, 1044], \quad [1044, 1046]$$

then cover $[855, 1046]$. $\square$

Since the five intervals in Lemma **??** overlap, they give

$$[215, 1046] \subseteq 5\big(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\big)_{\geq 12}.$$

Proposition 8. *Every integer* $n \geq 215$ *is a sum of exactly five elements of* $\big(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\big)_{\geq 12}$*. Consequently,*

$$[212, \infty) \subseteq F_5(\big(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\big)_{\geq 12}).$$

The lower endpoint is sharp: none of $209, 210, 211$ *belongs to* $F_5(\big(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\big)_{\geq 12})$*.*

Proof. Lemma **??** gives the stronger inclusion $[215, 1046] \subseteq 5\big(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\big)_{\geq 12}$; only the initial interval $[215, 867]$ is needed as the base case for strong induction on n. Let $n \geq 868$, and choose the unique $S \in \{5, 6, 7, 8\}$ with $S \equiv n \pmod 4$. Then

$$M = \frac{n - S}{4} \geq \frac{868 - 8}{4} = 215.$$

Since $M < n$, the induction hypothesis gives

$$M = q_1 + \cdots + q_5, \qquad q_i \in \big(\tfrac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\big)_{\geq 12}.$$

Because $5 \leq S \leq 8$, we may choose $e_i \in \{1, 2\}$ so that $e_1 + \cdots + e_5 = S$. Therefore

$$n = 4M + S = \sum_{i=1}^{5} (4q_i + e_i).$$

Lemma **??** shows that every summand $4q_i + e_i$ belongs to $\big(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\big)_{\geq 12}$. Thus $n \in 5\big(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\big)_{\geq 12}$, completing the induction.

Finally,

$$\begin{aligned} 212 &= 53 + 53 + 53 + 53, \\ 213 &= 53 + 53 + 53 + 54, \\ 214 &= 53 + 53 + 54 + 54, \end{aligned}$$

so $[212, \infty) \subseteq F_5(\big(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\big)_{\geq 12})$. We next verify that the only elements of $\big(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\big)_{\geq 12}$ below 212 are the eight members of L. Primitive Dyck words of lengths 4, 6, and 8 give, after division by 4, respectively,

$$\{3\}, \qquad \{13, 14\}, \qquad \{53, 54, 57, 58, 60\}.$$

Every primitive Dyck word of length 10 has the form $1u0$, where u is a Dyck word of length 8. The lexicographically least such word is 1101010100, whose value is 852; after division by 4 this gives 213. Longer primitive words have still larger values. Hence $\left(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\right)_{\geq 12}\cap[0,211]=L$.

A sum of at most five members of L using at most three terms from $\{53,54,57,58,60\}$ is at most

$$3\cdot 60+2\cdot 14=208,$$

whereas a sum using at least four such terms is at least

$$4\cdot 53=212.$$

Hence $209,210,211\notin F_5(\left(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\right)_{\geq 12})$. $\square$

4 The exceptional set

We first translate six-summand representability in the congruence class 2 modulo 4 into a finite-sum condition on $\left(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\right)_{\geq 12}$.

Lemma 9. *Let $N=4m+2$. Then N is a sum of at most six positive primitive Dyck numbers if and only if*

$$m\in F_5(\left(\tfrac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\right)_{\geq 12})\cup\left(1+F_3(\left(\tfrac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\right)_{\geq 12})\right)\cup\left(2+F_1(\left(\tfrac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\right)_{\geq 12})\right).$$

Proof. By Lemma **??**, every primitive Dyck summand other than 2 is a multiple of 4. Hence a representation of $N\equiv 2\pmod 4$ contains an odd number of copies of 2. If it has at most six summands, that number is 1, 3, or 5.

With one copy of 2, division of the remaining sum by 4 gives $m\in F_5(\left(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\right)_{\geq 12})$. With three copies it gives $m-1\in F_3(\left(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\right)_{\geq 12})$, and with five copies it gives $m-2\in F_1(\left(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\right)_{\geq 12})$. This proves necessity. Conversely, each of the three memberships gives a representation of N after multiplying the corresponding $\left(\frac{1}{4}\cdot\mathcal{N}_{\mathrm{pD}}\right)_{\geq 12}$-summands by 4 and adjoining, respectively, one, three, or five copies of 2. Thus the condition is also sufficient. $\square$

Lemma 10 (Small multiples of four)**.** *Among the positive multiples of* 4 *below* 100*, the only integer not representable by at most six positive primitive Dyck numbers is* 44*. Moreover,*

$$44=12+12+12+2+2+2+2.$$

Proof. The positive primitive Dyck numbers below 100 are precisely

$$2,\ 12,\ 52,\ 56.$$

First suppose $N < 52$ and write $N = 4x$. A sum of copies of 12 and pairs $2+2$ has the form

$$x = 3a + c$$

and uses $a + 2c$ summands. Under the condition $a + 2c \le 6$, the possible positive values of $x < 13$ are

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12;$$

the sole missing value is $x = 11$, corresponding to $N = 44$.

For $52 \le N < 100$, one first uses either $52 = 4 \cdot 13$ or $56 = 4 \cdot 14$. With at most five remaining summands, the residual quotients obtainable from 12 and pairs $2+2$ are

$$R = \{3a + c : a, c \ge 0,\ \ a + 2c \le 5\} = \{0, 1, 2, 3, 4, 5, 6, 7, 9, 10, 12, 15\}.$$

A direct inspection of these twelve integers gives

$$[13, 24] \subseteq (13 + R) \cup (14 + R),$$

so every multiple of 4 from 52 through 96 has the required representation. The displayed seven-term representation of 44 finishes the proof. $\square$

Lemma 11 (Finite exceptional-set certificate)**.** *Let*

$$L = \{3, 13, 14, 53, 54, 57, 58, 60\}.$$

Then

$$\begin{aligned} F_5(L) \cap [0, 211] = \{0, 3, 6, 9\} &\cup [12, 17] \cup [19, 20] \cup [22, 23] \\ &\cup [25, 37] \cup [39, 48] \cup [52, 208]. \end{aligned}$$

Moreover,

$$\begin{aligned} [0, 211] \cap \Big(F_5(L) \cup (1 + F_3(L)) \cup (2 + F_1(L))\Big) \\ = [0, 7] \cup [9, 10] \cup [12, 23] \cup [25, 37] \\ \cup [39, 48] \cup [52, 208]. \end{aligned}$$

Consequently, the complement in $[0, 211]$ *is*

$$\{8, 11, 24, 38, 49, 50, 51, 209, 210, 211\}.$$

Proof. Starting with $0L = \{0\}$ and using the recursion $jL = (j-1)L + L$, successive addition of the eight elements of L gives, up to 211,

$$\begin{aligned} F_5(L) \cap [0, 211] = \{0, 3, 6, 9\} &\cup [12, 17] \cup [19, 20] \cup [22, 23] \\ &\cup [25, 37] \cup [39, 48] \cup [52, 208], \end{aligned}$$

while

$$\begin{aligned} (1 + F_3(L)) \cap [0, 211] = \{1, 4, 7, 10\} &\cup [14, 15] \cup [17, 18] \cup [20, 21] \\ &\cup [27, 32] \cup [40, 43] \cup [54, 55] \cup [57, 62] \\ &\cup [64, 65] \cup [67, 78] \cup [80, 89] \cup [107, 135] \\ &\cup [160, 179] \cup \{181\}, \end{aligned}$$

and

$$(2 + F_1(L)) \cap [0, 211] = \{2, 5\} \cup [15, 16] \cup [55, 56] \cup [59, 60] \cup \{62\}.$$

Taking the union of these displayed interval lists gives the second formula, and its omitted integers are exactly those in the final set. Each step involves only translating the preceding interval list by the eight elements of L, so the calculation can be checked without an exhaustive search. $\square$

Proof of Theorem **??**. Suppose first that $N \equiv 0 \pmod 4$. If $N \geq 100$, write $N = 4n$. Then $n \geq 25$, so Proposition **??** and (**??**) give a representation of N with at most six primitive Dyck summands. Lemma **??** treats the remaining positive multiples of 4 and shows that $r(44) = 7$.

Now let $N \equiv 2 \pmod 4$, say $N = 4m + 2$. If $N \geq 850$, then $m \geq 212$, so Proposition **??** gives $m \in F_5(\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\right)_{\geq 12})$. By Lemma **??**, N is a sum of at most six primitive Dyck numbers. Together with the preceding paragraph, this already shows that every even $N \geq 848$ has such a representation.

It remains to classify the numbers $N \leq 846$ with $N \equiv 2 \pmod 4$. Here $0 \leq m \leq 211$, and Proposition **??** shows that the only elements of $\left(\frac{1}{4} \cdot \mathcal{N}_{\mathrm{pD}}\right)_{\geq 12}$ not exceeding m lie in

$$L = \{3, 13, 14, 53, 54, 57, 58, 60\}.$$

Lemma **??** shows that the values of m not covered by the criterion are

$$\{8, 11, 24, 38, 49, 50, 51, 209, 210, 211\}.$$

By Lemma **??**, the corresponding values $N = 4m + 2$ are precisely

$$34, 46, 98, 154, 198, 202, 206, 838, 842, 846.$$

Thus these ten numbers, together with 44, are exactly the positive even integers that are not sums of at most six primitive Dyck numbers.

All of them except 46 have seven-summand representations:

$$\begin{aligned}
34 &= 12 + 12 + 2 + 2 + 2 + 2 + 2, \\
44 &= 12 + 12 + 12 + 2 + 2 + 2 + 2, \\
98 &= 56 + 12 + 12 + 12 + 2 + 2 + 2, \\
154 &= 52 + 52 + 12 + 12 + 12 + 12 + 2, \\
198 &= 56 + 52 + 52 + 12 + 12 + 12 + 2, \\
202 &= 56 + 56 + 52 + 12 + 12 + 12 + 2, \\
206 &= 56 + 56 + 56 + 12 + 12 + 12 + 2, \\
838 &= 240 + 240 + 240 + 52 + 52 + 12 + 2, \\
842 &= 240 + 240 + 240 + 56 + 52 + 12 + 2, \\
846 &= 240 + 240 + 240 + 56 + 56 + 12 + 2.
\end{aligned}$$

Proposition **??** shows that 46 requires exactly eight summands. This proves all assertions of the theorem. □

Corollary 12. *The integer 848 is the least even integer T such that every even integer $N \geq T$ is a sum of at most six primitive Dyck numbers.*

Proof. Theorem **??** shows that six summands suffice for every even integer at least 848, whereas 846 requires seven summands. □

Corollary 13. *Every nonnegative even integer is a sum of at most eight elements of $\mathcal{N}_{\mathrm{pD}}$, and 46 is the unique even integer that is not a sum of at most seven elements of $\mathcal{N}_{\mathrm{pD}}$.*

Proof. The assertion for positive integers follows immediately from Theorem **??**; the integer 0 is the empty sum. □

Theorem **??** gives the complete exceptional set for six-summand representability. Corollary **??** shows that $\mathcal{N}_{\mathrm{pD}}$ is a relative additive basis of exact order eight for $2\mathbb{N}_0$. It is also a relative asymptotic additive basis of order at most six; equivalently, $\frac{1}{2} \cdot \mathcal{N}_{\mathrm{pD}}$ is an asymptotic additive basis of $\mathbb{N}_0$ of order at most six. Corollary **??** shows that the associated threshold 848 is sharp.

Let h_{asym} denote the least h for which every sufficiently large even integer is a sum of at most h primitive Dyck numbers. The larger family of all totally balanced numbers is not an asymptotic additive basis of order two [**?**, Thm. 14]; since the primitive family is a subset of it, two summands cannot suffice here either. Together with the present upper bound, this gives

$$3 \leq h_{\mathrm{asym}} \leq 6.$$

The exact asymptotic order is therefore one of $3, 4, 5, 6$. In particular, it remains open whether the bound six can be reduced to five.

5 Sequences from the OEIS

Three OEIS sequences are discussed in this paper. The totally balanced numbers form A014486 [**?**]. The positive elements of $\mathcal{N}_{\mathrm{pD}}$, the primitive Dyck numbers, form A057547 [**?**]. The central sequence studied here, $a(n) = r(2n)$, is A395858 [**?**].

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